

Nuclear Medicine Notes: Radioactive Activity

1 Activity Formula

The activity of a radioactive sample is

$$A = -\frac{dN}{dt}$$

or equivalently,

$$A = \lambda N$$

where

- A = activity,
- N = number of radioactive nuclei,
- λ = decay constant.

2 Example: Technetium-99m

Given a Technetium-99m sample with a half-life of

$$T_{1/2} = 6 \text{ h}$$

and

$$N = 1.0 \times 10^{15} \text{ radioactive nuclei.}$$

Step 1: Convert hours to seconds

$$6 \text{ h} \left(\frac{3600 \text{ s}}{1 \text{ h}} \right) = 21600 \text{ s}$$

Step 2: Calculate the decay constant

Using

$$\lambda = \frac{\ln 2}{T_{1/2}},$$

we get

$$\lambda = \frac{\ln 2}{21600 \text{ s}} = 3.21 \times 10^{-5} \text{ s}^{-1}.$$

Step 3: Calculate the activity

$$A = \lambda N$$

$$A = (3.21 \times 10^{-5} \text{ s}^{-1})(1.0 \times 10^{15})$$

$$A = 3.21 \times 10^{10} \text{ Bq}$$

3 Curie and Radioactivity Notes

- Curie discovered radium and polonium.
- The term “radioactivity” was coined by Marie Curie.
- One curie (Ci) is

$$1 \text{ Ci} = 3.70 \times 10^{10} \text{ disintegrations per second (dps)}.$$

- The handwritten source also notes that this corresponds to the activity associated with approximately 1 g of radium-226.